17 Kelly grows potatoes.
She has 2500 kg of potatoes, correct to the nearest 10 kg.

Kelly is going to put the potatoes into sacks.
Each sack will contain 12.5 kg of potatoes.

Kelly assumes that 12.5 kg is measured correct to the nearest 0.1 kg.

(a) Does Kelly definitely have enough potatoes to fill 200 sacks?
You must show how you get your answer.

Potatoes: 2495 ≤ weight < 2505 kg

1 mark
Use correct rounding bounds.

Sack: 12.45 ≤ weight < 12.55 kg

Near the upper bound, 200 sacks need almost

1 mark
Compare worst case weights.

200 × 12.55 = 2510 kg, which exceeds 2495 kg.

No: there is no guarantee of enough potatoes.

1 mark
No, supported by the bounds.

(3)

Kelly's assumption is wrong.
12.5 kg is measured correct to the nearest 0.5 kg.

(b) Explain how this could affect your decision in part (a).

Still not definite: the sack upper bound rises to 12.75 kg.

The possible requirement rises towards 2550 kg > 2495 kg.

1 mark
Wider uncertainty; decision unchanged.

(1)

(Total for Question 17 is 4 marks)